

Pierre Moreau

PhD in Computer Science · Data Scientist / ML Engineer — AI & Healthcare

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Summary

PhD in Computer Science (AI / Machine Learning) specialised in healthcare, bridging applied research and hands-on engineering — from model development to deployment. Skilled in generative AI (LLMs, RAG), multimodal learning and shipping full-stack AI products (FastAPI / Django / Next.js, Docker, self-hosted). I build interpretable, reliable AI with real-world impact.

Skills

Languages: Python, SQL, JavaScript, Bash, PHP, HTML/CSS

ML & Deep Learning: PyTorch, TensorFlow, Keras, Hugging Face Transformers, scikit-learn, XGBoost

LLMs & Generative AI: Fine-tuning (PEFT, LoRA), RAG (LangChain), prompting, Whisper (speech-to-text), OCR

Vision & NLP: Segmentation (UNetR, SwinUNet), OpenCV, Pillow, NLTK

Data Science: NumPy, Pandas, SciPy, Matplotlib, Seaborn ; PCA, DTW, ANOVA, Chi²

Engineering & Full-stack: Django, FastAPI, Flask, Next.js / React, TypeScript, PostgreSQL / Supabase, Stripe

MLOps & Deployment: Docker, Git / GitHub, REST APIs, self-hosting (Linux / DDNS), GitHub Pages, Google Play Console

Responsible AI: Explainability (SHAP, LIME), bias/fairness analysis, GDPR & AI Act

Tools: Git, Docker, VS Code / Cursor, Jupyter, Claude Code / Codex

Professional Experience

Post-doctoral Researcher — AI & Healthcare, University of Picardie Jules Verne – Amiens Sept 2025 – present

- RAUC project: designing an optimised care pathway for digestive surgical emergencies, integrating AI and e-health, aimed at predicting post-operative complications.

Data & AI Engineer — AI & Data Science Lead (Healthcare), Capgemini Engineering – Paris Mar 2024 – May 2025

- Designed an AI-based medical assistant demonstrator with multimodal workflows.
- Built NLP pipelines: OCR of blood test results, speech-to-text transcription (Whisper), automatic keyword extraction to enrich metadata.
- Fine-tuned LLMs (OpenAI, Mistral) with Hugging Face Transformers and PEFT for clinical summaries and audio transcription.
- Developed a multimodal RAG pipeline (LangChain) synthesising structured and unstructured medical data.
- Applied research on placenta segmentation (UNetR, SwinUNet) to improve accuracy in prenatal imaging.
- Coordinated AI researchers, data scientists and clinicians; project and team management (15-20 people).

Teaching & Research Assistant (ATER), University of Picardie Jules Verne – Amiens Sept 2022 – Aug 2023

- Tutorials and lab sessions for undergraduates (years 1-3) — PHP, Bash, C, advanced programming.

PhD Candidate — Doctoral Research, University of Picardie Jules Verne – Amiens Sept 2019 – June 2023

- Research on human motion modelling and analysis (in collaboration with Amiens University Hospital); built a comprehensive motion database.
- Analysis (PCA, Dynamic Time Warping, correlation) and optimisation via genetic algorithms for feature selection.
- Deep learning modelling (RNN, LSTM, CNN, residual networks, FcNN) and SVM; interpretable biomechanical representation using matrix formulations.

IT Engineer / Developer (apprenticeship then permanent), LV Informatique – Sept 2017 – Sept 2019
Amiens

- Developed custom ERP software (Lynéade then Lynéa2, WinDev): business processes (accounting, purchasing, sales, inventory, after-sales), SQL database administration, client support and customisation.

Projects

Solis — AI-powered ERP (independent product) 2024 – present

End-to-end full-stack ERP with a conversational AI assistant; solo-designed, built and self-hosted.

- Backend Django REST + PostgreSQL / Supabase, frontend Next.js 15; containerised with Docker, self-hosted (Linux).
- AI features: Claude API assistant, Whisper speech-to-text, OCR document scanning.
- Business logic: quotes / invoices, e-invoicing (Factur-X), Stripe billing; GDPR-compliant by design.

Vesta — radiology scheduling (Amiens University Hospital) 2024

Web app used by a radiologist to auto-generate the department's monthly planning; React / Vite / TypeScript, deployed on GitHub Pages.

ReapIt — AI strength-training companion (mobile) 2025

Mobile app that generates and adapts personalised strength-training programs with AI; shipped in closed beta on Google Play.

Education

University of Picardie Jules Verne, PhD in Computer Science — Artificial Intelligence Sept 2019 – June 2023
– Amiens

- Thesis: analysis of human motion using artificial intelligence tools, in collaboration with Amiens University Hospital.

University of Picardie Jules Verne, MSc in Computer Methods Applied to Business Sept 2017 – June 2019
Management (MIAGE) – Amiens

University of Picardie Jules Verne, BSc in Computer Methods Applied to Business Sept 2014 – June 2017
Management (MIAGE) – Amiens

Publications

A Genetic Algorithm-Optimized PID Controller for Robust Tumor Treatment via Combined Immuno-Chemotherapy 2026

Jérôme Bosche, Consibelle Djiffouo Dzoyim, *Pierre Moreau*
(International Conference on Control, Decision and Information Technologies (CoDIT) — accepted)

Towards Better Placenta Segmentation for Earlier Preeclampsia Detection on 3D Doppler Images 2025

Charlotte Alliod, Emilien Micard, *Pierre Moreau*, Léo Challier, et al.
(Computerized Medical Imaging and Graphics (CMIG) — submitted)

Two-branch neural network using two data types for human activity recognition 2023

Pierre Moreau, David Durand, Jérôme Bosche, Michel Lefranc
(IEEE Sensors Journal)

A motion recognition technique based on linear matrix representation to improve Parkinson's disease treatments 2021

Pierre Moreau, David Durand, Jérôme Bosche, Michel Lefranc
(9th International Conference on Systems and Control (ICSC))

A motion recognition algorithm using polytopic modeling 2020

Pierre Moreau, David Durand, Jérôme Bosche, Michel Lefranc
(7th International Conference on Control, Decision and Information Technologies (CoDIT))

Certifications

Generative AI with Large Language Models — DeepLearning.AI & AWS (2024)

Building Multimodal Search and RAG — DeepLearning.AI (2025)